

v3 defringe pipeline

Writes <channel>/defringe_v3/ only. Does not overwrite defringe_v22. Catalog is not appended.

1. Raw stack · ChanA_stk.tif / ChanB_stk.tif
Never overwritten. All v3 products sit beside it in defringe_v3/.

2. Shutter detect — FOV std cliff (not mean; PMT offset stays ~850/738 ADU)
Quiet if $\text{std} \leq 0.40 \times p75(\text{std})$ for ≥ 3 frames after a drop $\geq 0.35 \times \text{live std}$.
Logged on overview + families.json. Not a hard seed: image-test still vetoes shutter q on live.

3a. FFT-domain proposals

- Catalog A/B/C (q / fx support only)
- Same-raster cache prior
- Shutter-learn fy (incomplete)
- fy-row detect on leftover

3b. Image-domain proposals

- Congruence: H, V, both diagonals
- One (qy, qx) or none
- fx and fy stay separate cand
- Never max(fy,fx) into one gate

4. Image-test leftover · eval = shutter u anchors 160 / 700 / 1061
Order: catalog → shutter-learn → linescan → leftover FFT. Notch one candidate. Score removed (coverage, evenness, ridges). PASS → keep family, leftover that frame. FAIL → discard (PDF).
Do not retune a rejected mask. Repeat ≤ 4 rounds. Two consecutive rejects stop.

5. Track kept families · q walk ± 10 bins / 50-frame block · apply search ± 2
fx/fy support is the seed template and does not wander. Same-axis families cannot hop onto each other.

6. Union apply (pack_D) per frame · one FFT → each family on its own axis → one IFFT
Outputs: *_defringed_v3.tif · *_removed_v3.tif · per_frame.csv · overview.pdf

Leftover loop and apply — what is allowed to move

Proposal uniqueness is (axis, q), not q alone. fy q=10 and fx q=10 can both be kept.
A failed family is not reshaped. Recursion means: evaluate the next unused candidate on leftover.

Keepers are union-applied. Gate and local column-excess scale how hard a family is notched;
the applied heatmap can shrink inside the seed fx/fy support. q may drift a few bins from seed.

Outputs under <channel>/defringe_v3/

- *_defringed_v3.tif cleaned stack (same dtype as raw)
- *_removed_v3.tif raw — cleaned (float32)
- per_frame.csv q / gate / drift / axis / source per family
- families.json shutter, leftover rounds, kept families, summary
- mean_raw / mean_cleaned / mean_removed.tif
- overview.pdf shutter, metrics vs frame, discarded masks, inspect samples

Inspect pages pick: shutter mid, anchors 160/700/1061, strongest-gate linescan vs FFT-seed frames, drifted-q frames ($|q - q_{\text{seed}}| \geq 2$ while gated), and strongest RMS. Each sample shows the four linescans + their mask contributions, FFT-family seed masks, the final applied mask, then original / predicted fringe / actual removed / cleaned.